

Drone Research Proposal

Davit Komyakov, 11/27/2024

Developing an understanding of wildlife migration is crucial for the conservation and management of the ecosystem. There are many environmental factors influencing the migration of wildlife, however, recorded research on the matter is very limited. Therefore, the aim of this proposal is to analyze the migration patterns of deer in the wild through the use of drones.

The research question that the drone investigation will aim to answer is: How can quantitative and qualitative data be collected on Migration patterns of deer through all seasons? By answering this question, it will provide a better understanding of the wildlife in certain regions similar to that of Banff and aid in efforts to conserve the environment as best as possible.

By classifying this research proposal as a medium-scale project, the setting of the proposed investigation will be in a 30 square kilometer area of Banff National Park. The exact area of the environment is determined by the battery life of each rotation of the drones, around 20 to 45 minutes, as well as the average distance a drone can cover, with an estimated average of 15 kilometers. Moreover, the evaluation of locations took into consideration having a diverse and rich vegetation environment, as well as separation from urban society to ensure minimal human interference to provide the most ideal conditions for observing natural migration patterns.

The research timeline will span one whole calendar year to analyze migration patterns for all four seasons and avoid biases in the data collection. To collect data that can be used to train machine models, it is necessary to include all seasons as the environment of the deer changes, thus influencing their habits and migration patterns. Because, if this data was used to train a machine learning model, not including all seasons, therefore, all migration patterns of deer, there is potential to introduce bias and reduce the model's prediction accuracy, as the model would not account for seasonal variability. As for the number of drones, 20 drones is sufficient to scan significant amounts of the area, each with its own designated sub-area to

analyze, while also having backup drones to cover for drones that need to charge before their next cycle.

As for the equipment, the drones will be equipped with thermal cameras to capture the heat signatures of deer and their markings to track their movements after they have left the location. Or in other cases, the thermal cameras will improve data collection at night, or in densely vegetated areas of the park where deer would otherwise be difficult to locate. Alongside this, the drones will be mounted on LiDar sensors that can more exactly determine the displacement of the location of the deer, leading to more accurate data collection. For a more in-depth explanation, Lidar sensors use light in the form of pulse lasers to measure angles and distortions on the surface to retrieve precise three-dimensional scans of the environment (NOAA, 2023). In doing so, it will help researchers analyzing the drone's footage determine patterns in environmental factors or to detect new ones.

Before the project is initiated, it is crucial to adhere to the national parks' laws and regulations as it is important to maintain ethics while researching. The custom drones equipped with heat vision cameras will weigh over 250 grams, the Canadian government's official website states that any drone above 250 grams and below 25 kilograms must be registered. Each drone utilized in the experiment must be given a registration number as well. (Government of Canada, 2023). Furthermore, actions such as minimizing disturbance to Banff wildlife and maintaining data privacy should remain a priority for ensuring that the research does not interfere with nor affect the wildlife, or violate privacy laws and maintain ethics.

After concluding the experiment, the data collected from the thermal cameras, LiDar sensors, as well as time series observations will provide both qualitative and quantitative insights into the seasonal migration patterns of the deer. Researchers of the project will then have the opportunity to integrate these insights into other data collection sources such as satellite imagery and weather data to fully develop an understanding of deer migration patterns. These new findings and better understanding will in turn help researchers with future projects

involving migration patterns by integrating the data with machine learning, to be able to predict outcomes and give rough estimates of the resulting data from the experiments. For guidance and inspiration, researchers experimenting can relate to a previous study involving drones scanning the environment to form a better understanding of a matter by DroneHive. In the experiment, DroneHive equipped a drone with a LiDar sensor and multispectral cameras to scan a vineyard, creating detailed models of the environment and developing a better understanding of what outside factors are beneficial to plant health (DroneHive, 2017). Another, by incorporating the data from the experiment into a machine learning model, it can identify the relationship patterns between deer and the environment and provide support for future research.

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